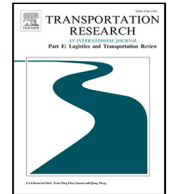




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Operational benefits of autonomous vessels in logistics—A case of autonomous water-taxis in Bergen

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ABSTRACT

Although the technology of autonomous vessels has developed rapidly in the past decade, the literature on the effects of this new technology on logistics and transportation is limited. Particularly, the potential operational changes brought by the autonomy of vessels are not discussed. Besides savings on crew, other benefits of autonomy, especially from an operational perspective, remain unclear. On the other hand, countries like Norway are facing significant challenges in their urban and rural logistics. The land-based public transportation system cannot efficiently serve the demands, especially for areas isolated or separated by fjords or rivers. Water-based transportation systems are thus very attractive, and autonomy makes such systems economically more viable. In this paper we focus on a potential case of autonomous water-taxis in Bergen, Norway. First, we identify the major operational differences, from a logistics perspective, between conventional manned water-taxis and autonomous ones. Then a mixed integer linear programming model is developed to tackle the facility location, fleet allocation and routing decisions in the daily operations of a water-taxi business. The model in this paper is used to analyse the problem, rather than designed for direct use during implementation. In the computational study, different cases with either conventional or autonomous water-taxis are tested. The results show that autonomy can lead to substantial savings on travelled distance due to number and locations of parking terminals being feasible only for autonomous systems.

1. Introduction

Due to the development of artificial intelligence (AI) and Information and communications technology (ICT), the potentials of autonomous technology in urban logistics systems start to attract more attention. The technologies of autonomous cars and autopilots on the roads are quite mature. Hence important players in the industry, such as Google and Tesla, have started to explore the commercialization of autonomous technology in land-based transportation systems (Tesla, 2018; Waymo, 2019). Correspondingly, future applications of autonomous vehicles in city logistics are also widely discussed in the literature. Cao and Ceder (2019) developed an optimization model to determine the optimal timetable for autonomous shuttle buses based on real-time passenger demand. Dai et al. (2020) studied the joint optimization problem for a mixed fleet with both autonomous and manned buses in terms of their scheduling and capacity planning. Liang et al. (2016) proposed two integer programming models to optimize the service area and trip selection of autonomous taxis for the last mile of train trips. Liang et al. (2017) developed a profit maximization model for the vehicle routing problem of autonomous taxi trips with dynamic travel times. Liang et al. (2020) tackled the routing problem

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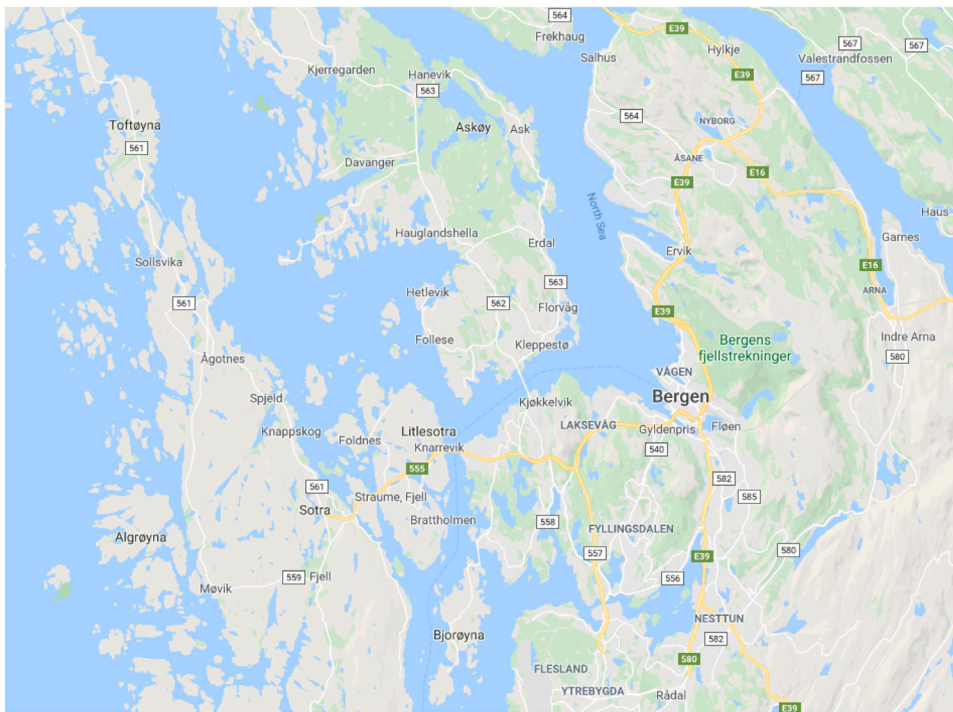


Fig. 1. Map of Bergen area.

of autonomous taxis with ride-sharing and congestion-based travel times. Other papers regarding the application of autonomous buses or taxis can be found in [Abe \(2019\)](#), [Hörl \(2017\)](#), [Duan et al. \(2020\)](#)

For autonomous vessels, however, the focus of the studies still remains on the technical level. [Gu et al. \(2020\)](#) show that although the number of publications regarding autonomous or unmanned vessels has increased rapidly in the last decade, about two-thirds of the papers are still discussing the fundamental technical details, such as trajectory planning and collision avoidance, which are vital for the navigation of autonomous vessels. [Akbar et al. \(2020\)](#) discussed the potential operational benefit of adopting autonomous container ships in a short-sea liner shipping network. They assumed a mixed fleet operation with conventional mother ships and autonomous daughter ships performing cargo transshipment at specific ports. However, similar research about applying autonomous vessels in logistics and transportation is still limited. One potential reason for such a difference between research on autonomous vehicles and vessels is that the technological development of autonomous vessels is relatively slower than for autonomous cars. Nevertheless, recent new projects initiated by the shipping industry, for example [Kongsberg \(2018\)](#) and [Rolls-Royce \(2018\)](#), reflect that the confidence on the maturity of this technology is increasing. Therefore, more efforts from an academic perspective should also be invested to explore the potentials of autonomous vessels in our logistics systems. We focus on understanding ways in which operational characteristics of transport systems will change as a result of the new technology, thereby expanding the overall picture of what autonomy brings to waterborne transportation.

Norway is a country with a complex geographical landscape. More than 1000 fjords in Norway significantly fragmentize the coastal areas of the country, which brings great challenges to the logistics system that connects these isolated lands and may eventually lead to imbalanced developments among different regions. Bergen, the second largest city in Norway, faces similar problems. The intensive network of waterways shown in [Fig. 1](#) dramatically affects the efficiency of passenger and cargo transportation. Moreover, due to an increasing number of cars, city logistics also faces other kinds of challenges including road congestion, greenhouse gas emissions and noise pollution. Water-based transport is certainly an alternative to overcome the problems of low transport efficiency and environmental pollution. Nevertheless, given the population and demand for transport in this region, it is very unlikely to have economies of scale to facilitate a water-based city logistics system due to high labour costs; Both passengers and cargo need to adapt to the existing land-based transport. This leads to rather long and time consuming travel between many locations in the Bergen area, even though the locations are actually very close if you can go by water. With the help of autonomy, water-based transport services will become more competitive since the high labour costs will no longer be an issue. Therefore, there is a very strong motivation, from both academic and social perspectives, to explore the potential benefits of autonomous vessels in city logistics in the Bergen area. In this paper, we focus on the operation of a water-taxi fleet which offers transport service for customers with higher priority on travel time. Ride-sharing is therefore not considered.

The most widely recognized advantage of autonomous technology is the reduction of labour cost. The disadvantages of autonomous vessels are also obvious. High initial investment and increased legal and liability complexity are often mentioned in

the literature. However, the focus of this paper is not the trade-off between the increased initial investment in technology and the decreased variable cost due to the removal of staff. Readers who are interested in the cost–benefit analysis of autonomous vessels may find reference in [Kretschmann et al. \(2017\)](#) and [Hogg and Ghosh \(2016\)](#).

The purpose of this paper is, however, to explore what else we can expect from the autonomy of vessels besides the obvious cost savings on salaries. For example, since the comfort and safety of water-taxi drivers are no longer an issue, unprecedented locations, such as an isolated peninsula or a small island, become possible terminals for autonomous water-taxis. Note that the security concerns regarding theft and damage will still exist even if autonomous technology is adopted. But such concerns will be handled by remote monitoring. Moreover, since the operation of autonomous vessels will heavily rely on the internet, cyber security issues, such as hacking, become critical. However, as for autonomous cars, autonomous water-taxis will have certain mechanisms to allow such as a remote monitoring centre to take over the control and stop the vessel when such an emergency happens. More details on measures to tackle cyber security concerns can be found in [Cheung et al. \(2021\)](#). On the other hand, it is likely that the cost of a terminal for autonomous water-taxis will be decreased because all the human-related facilities are no longer needed. Hence, additional terminals maybe introduced if autonomous water-taxis are adopted. Last, since the terminals are also autonomous or remote-monitored, it is possible to change the number and location of the open terminals and redistribute the water-taxi fleet in different time periods as long as the infrastructures are built. It is necessary for the terminals to be autonomous or remote-monitored. Otherwise, the feasibility of multiple terminal decreases and the isolated locations become infeasible due to costs related to human factors.

We do not provide a comprehensive analysis including all the pros and cons of autonomous vessels. This study does not aim to prove neither the superiority nor the inferiority of autonomous technology compared to conventional ones, since many factors related to engineering and security, as well as investment costs, are discussed elsewhere in the literature, and are in any case not yet established. We choose to focus on the logistics using autonomous vessels, since that part of the overall picture is still not well studied. In general, we study the potential differences between autonomous and conventional manned vessels from a daily operations perspective and the corresponding benefits. We ask the following question: In the future, will we use autonomous vessels the same way we use conventional vessels now? Alternatively, will this new technology bring significant changes to the operations of transport and logistics? As a first step, we choose to focus on the case of autonomous water-taxi for urban transport in Bergen.

Strategic and operational decisions are tightly connected. Strategic decisions set boundaries for operational decisions, but more importantly, operational decisions affect what are good strategic decisions, which is the same case in this study. So to understand the potential value of autonomy on water taxis, we need to model both strategic and operational decisions together, despite the fact that they have very different time scales. This is why this paper, which is about operational possibilities that emerge from autonomy is analysed in a model that seems to focus on strategic decisions. The value of the new operational features can only be seen in a context where the strategic decisions fit the operational possibilities.

There are three major contributions in this study. First of all, the paper discusses and explores potential operational advantages of autonomous vessels, apart from the well-established labor cost savings. These advantages includes unprecedented terminal locations, multi-terminal feasibility and increased routing flexibility. Second, a new mixed-integer linear programming model for the facility location, fleet allocation and routing problem of autonomous water-taxis is developed. Last, a comprehensive computational study based on a realistic case is designed and conducted. Managerial insights about the operations of autonomous water-taxis are obtained. These insights are valid for similar cases around the world, as well as other autonomous water-based services.

The model developed in this study can be categorized as a variation of the classical location-routing problem (LRP) which has been widely studied in the literature. For more details regarding LRP, please check [Drexel and Schneider \(2015\)](#). The main novelties of our model compared to the existing literature are threefold. First, most LRP models consider loop routes in their decision-making process. This means that the vehicle has to return to the departing depot after visiting all the pickup and delivery points, see for example [Yu et al. \(2020\)](#), [Almouhanna et al. \(2020\)](#). In our model, the water-taxi is allowed to choose non-loop routes and return to different depots which significantly increases the routing complexity. Second, some existing works, for instance [Melkote and Daskin \(2001\)](#), have already documented the potential benefits of extra facilities. However, our model adopts not only the setting of multiple depots but also the possibility to change the number and location of the open depots during the operation. This new feature may not be possible if the depots are manned since the working hours of staff at different locations may not be so flexible. For autonomous or remote-monitored depots, it is not a problem since the staff who is monitoring different depots at different times could be the same. Last, the allocation of the fleet among multiple depots is often decided in the very beginning and cannot be changed afterwards. One exception is the bike-sharing problem in which the bikes are repositioned regularly ([Bulhões et al., 2018](#); [Lu et al., 2020](#)). The relocation of the bikes are performed by a certain number of vehicles which collect the bikes first and then redistribute them to different terminals. Such repositioning of bikes is passive as is not initiated by the bikes. In this paper, it is also possible to redistribute the water-taxi fleet during the day. But different from the repositioning of shared bikes, the relocation of the water-taxi fleet is actively carried out by the vessels themselves. The repositioning of the water-taxi is part of the routing decisions and hence no extra transport capacity is needed like in the bike-sharing problem.

However, we also need to clarify that the model in this paper is mainly used to study operational differences between conventional and autonomous water-taxi. It is not meant for direct real world applications. The model together with different scenarios are tested in the computational study so as to verify the operational advantages of autonomous vessels. The target readers of this study are stakeholders, such as researchers, transport service providers and local authorities, who are interested in the development and future application of autonomous vessels. This paper points out that there are more advantages than lower salary costs if autonomous vessels are adopted. The insights presented in this paper are not exclusive for the test location, namely Bergen, but can be generalized and applied to in other places with intense waterways. Moreover, the results are not only valid

for water-taxi services, but also meaningful for other potentially waterborne transport services in urban areas, such as package delivery and garbage collecting. We believe the model developed in this study is detailed enough, given a correct description of the topography, to learn about the operational advantages of the new technology which will be a critical reference for the policy makers in the future.

The paper is organized as follows. In Section 2, we describe the research problem. In Section 3, we present the formulation of the mathematical model and the corresponding explanations. Section 4 introduces the detailed settings and corresponding assumptions of the case study. In Section 5, the results of the computational study are illustrated and analysed. Finally, we conclude the paper in Section 6.

2. Problem description

In our study we assume that a city is divided by a fjord, river or lake, see Fig. 2. There is substantial demands for water-based transport among different key points of the city, for example the residential area (Point C) and the office area (Point A). The alternative connection with land-based transportation is much longer both in distance and travel time. The demands are clustered in different peak hours during a day. For instance, most of the water-based transport demands in the morning happen during the hours before work, typically 6 to 9 am. Such a cluster of demands is called a demand wave and there are multiple demand waves in a day. In one demand wave, different passengers have different origins and destinations as well as different departure and arrival times. In this paper, one single demand refers to one transport request for one or multiple passengers. Demand with a passenger group larger than the capacity of the water-taxi will be naturally considered as two or more demands with the same origin, destination and departure time.

A taxi company owning a fleet of water-taxis needs to establish a daily operation to serve the demand. Different from taxi services on land, it is very unlikely for a passenger to walk to the shore and just wait for a water-taxi to show up without booking or pre-checking the availability of the water-taxi at a certain location through mobile Apps. It is also impractical for a water-taxi to continuously patrol along the shore to seek random customers due to parking constraints and fuel costs. Therefore, we assume all passengers using the water-taxi service book in advance, including “I need it now”. The decisions involved in such an operation can be roughly divided in two.

The first decision made by the water-taxi company is the location of the “terminals” and “docking points”. A water-taxi terminal refers to a location with necessary infrastructure and facilities to allow the water-taxi to park for a longer period of time while waiting for the next customer. The main difference between terminal designs for regular taxis and water-taxis is certainly the parking facilities. A regular taxi can find many parking options, while a water-taxi has to return to a terminal for long-time berthing. Furthermore, if the water-taxi is electric, charging of the vessel will take place at a terminal while the water-taxi is waiting for new customers. Hence, charging facilities will also be needed at the terminals in this case. Since the average travelling distance per customer in Bergen is quite short, the frequency by which a water-taxi visits a terminal is rather high, compared to a regular taxi. Given the battery capacity of today’s electric cars (several hundred kilometres), we assume that charging will not be an issue when charging facilities exist in terminals. If autonomous water-taxis are adopted, a monitoring system is needed at the terminal to ensure the vessels will not be damaged or stolen. Moreover, special parking slots will be built to prevent the autonomous water-taxis from drifting away since no one will be there to tie the vessels to the dock. For basic maintenance, service personnel will be sent to the terminal when service is needed. For major repairs or hull cleaning, just as for conventional vessels, the service for autonomous water-taxis will be carried out at the service provider’s facilities. For manned water-taxis, a building is needed to allow a certain number of drivers to rest simultaneously. Hence, facilities including heating, electricity and toilet are required.

A docking point, on the other hand, refers to a location with limited infrastructure and facilities, probably just a simple quay with some lightning, so that the water-taxi can park there temporarily when it needs to pick up or unload a passenger. A water-taxi can also continue to park at the docking point after unloading the previous customer, if it knows that another passenger will show up shortly, for example within 10 min, according to the booking information at this location. But long-term parking is not permitted at any docking point due to the limited facilities. For instance, if the next passenger will arrive at the docking point after a relatively long or unknown amount of time, then the water-taxi cannot wait at the docking point and has to return to a terminal for long-term parking. The cost of a water-taxi terminal is therefore much higher than the cost of a docking point due to extra infrastructure and facilities. Normally, the number of water-taxi terminals should also be considered as a decision variable in the problem. However, since the autonomous water-taxi is still just a concept, the cost of a terminal is not known. Therefore, the number of terminals for the autonomous case will be treated as a parameter in this paper and different values will be tested and compared in Section 5.

The second layer of decisions for the water-taxi operation is the routing decision for each taxi vessel in the fleet. The entire operational time period in the problem is discretized into a number of time slots in which demands for transport among different locations emerge. The required departure and arrival time for each customer are also specified based on the time slots. The routing decisions for the water-taxis, including the choice of specific routes and the corresponding starting and ending time slots, make sure all passengers are served on time. For security concerns, we assume all idle water-taxis must return to a parking slot inside one of the terminals for long-term parking so that the vessel will not drift away and cause accident.

However, these decisions, namely terminal location and vessel routing, can be rather different for the autonomous case compared to the manned setup. For example, more terminals, unconventional terminal locations and new routing options become feasible with autonomous water-taxis. The main purpose of this paper is therefore to study the possible operational differences and the potential operational gains brought by autonomous water-taxis. As pointed out, this study needs to take both strategic and operational decisions into account, so that the strategic decisions are not counteracting the operational advantages of autonomy.

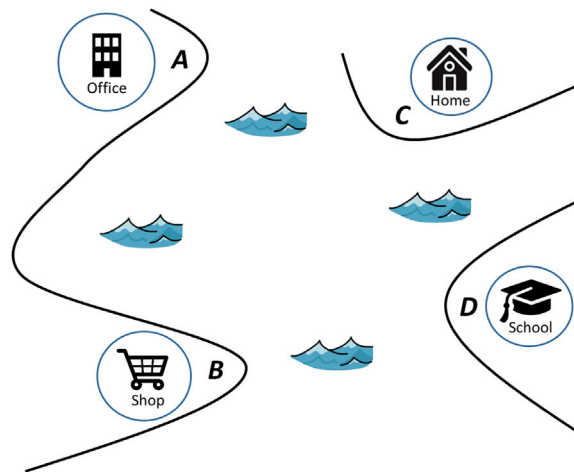


Fig. 2. Illustration of example city divided by water.

For conventional vessels, the parking of the water-taxi will be a big issue when a driver is waiting for the next customer. A taxi car on land may easily find a parking spot in the city, for instance at a gas station or near a convenience store, but a water-taxi has more restrictions regarding parking locations. First of all, you cannot let a manned water-taxi park in certain unusual locations, such as an isolated peninsula or a small island, for an unknown amount of time waiting for new demand. It will lead to both safety concerns and working condition problems for the drivers of the water-taxi since the weather may change quickly and the drivers need resting facilities which may not be available at these remote locations. Second, the water-taxi cannot park casually along the shore as well. The depth of the water, the draught of the vessel and the accessibility of docks need to be considered. Last, even if there is a simple dock, the limited parking capacity and the lack of infrastructure, such as resting facilities, will restrict the dock to only support temporary parking of the water-taxi when loading or unloading passengers. Therefore, in order to have a location where the water-taxi can be parked for a long time while waiting for new demands, a terminal is needed. A comprehensive water-taxi terminal creating not only parking places but also comfortable environment for drivers is mandatory. Such a terminal can be very expensive to build. According to Boligfiks (2020), the construction cost in Norway ranges from \$ 2,600 to \$ 5,300 per square metre. Hence, to build a 100 square metre terminal for manned water-taxi, will cost on average more than \$ 330,000 plus the land cost, which can be substantial in a city. Therefore the number of terminals for manned water-taxis will be very limited. In this paper we assume that only one comprehensive water-taxi terminal will be built if conventional vessels are used and the rest of the locations in the city will only have simple docking points for temporary parking. Hence, the water-taxi problem with conventional vessels can be viewed as a facility location problem. The key decision that needs to be made is then the location of the water-taxi terminal which minimizes the total travelling distance with the given fleet and demand. Note that demand is assumed deterministic, for example booked in advance, and that the taxi fleet is large enough to match the demand. If the situation is different, for instance, that it is not mandatory to satisfy demand and the size of the taxi fleet is limited, then the objective may become profit maximization. To choose an objective for the optimization model depends on the actual relationship between demand and supply, and the policy of the company.

For illustration, Fig. 3 shows a feasible solution in which the water-taxi terminal is built near the residential area (Point C) and docks for temporary parking are available in other areas (Point A, B, D). Once the location of the water-taxi terminal is decided, the routes for different passengers are also fixed. The routes in this case are all loop routes since, as mentioned previously, the water-taxi cannot park arbitrarily after serving the assigned passengers and thus needs to return to the only terminal for long-term parking and wait for the next customer. For example, to transport Passenger 1 from the office area (Point A) to the shopping area (Point B), a manned water-taxi will first be sent from the terminal to the office area to pick up this passenger. Then the passenger will be carried by the taxi to his or her destination at the shopping area. Then the water-taxi will temporarily park at the docking point B since Passenger 2 needs the service as well and he or she is already waiting there or will show up very soon. After unloading Passenger 2 at the school area, the water-taxi will not stay at docking point D since no passenger is waiting or showing up in a short time. Therefore, the water-taxi will return to the terminal for long-term parking and wait for the next customer.

For autonomous vessels, on the other hand, the flexibility of parking location is much higher. First, since the restrictions caused by human factors, e.g., safety and comfort concerns of the taxi driver, are removed by the autonomy of the vessel, the autonomous water-taxi can actually park at a remote terminal far away from the shore, for instance a small island in the middle of the water, for a long period of time. This new characteristic of autonomous vessels brings new alternatives of parking location for the water-taxi operation, for instance Point E in Fig. 4, which is not possible for conventional manned vessels. Such a terminal at a remote location can significantly decrease the total ballast voyages when the water-taxis serve the demands widely distributed among different areas in the city. Certainly, one may argue that it is possible to have a terminal for manned water-taxi in a remote location too. However, the construction costs for a manned terminal in a remote location will easily be enormous. Furthermore, the drivers may be isolated

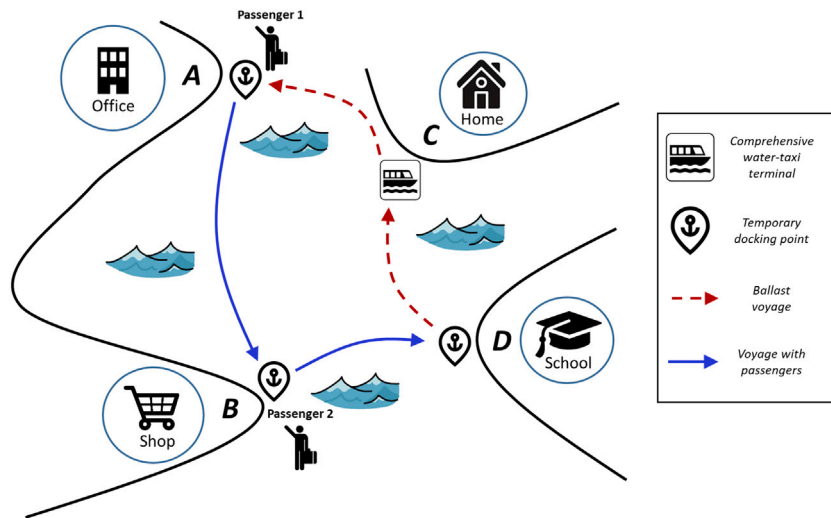


Fig. 3. Illustration of feasible solution with conventional water-taxi.

on the island during bad weather, which leads to considerable safety risks and inconvenience. Hence, a manned terminal in a remote location may be theoretically possible, but not practically feasible or realistic.

Second, the cost of the parking terminal for autonomous water-taxis will also be substantially lower due to the reduction of human related investment. Certainly, additional services are needed for such unmanned terminals, which will lead to extra costs. However, since the water-taxis are autonomous and a remote navigation function is unnecessary, the main service needed is remote monitoring of the terminals for security concerns. Therefore, the technology requirements is much lower, which leads to a limited initial investment. Moreover, the workload needed for monitoring a few terminals will be very limited. Hence, the water-taxi company may not need to build and operate its own control centre. The monitoring service is more likely to be outsourced to a third party remote control centre and use only a small fraction of its total capacity. Due to the economies of scale at the third party remote control centre, the additional variable cost, namely the labour cost for terminal monitoring, may also be limited. Therefore, we argue that in general the parking terminal for autonomous water-taxis will have a lower cost. Due to the lower cost, it is therefore possible to have multiple parking terminals if autonomous water-taxis are adopted, which dramatically increases the flexibility in routing decisions for the daily operations of the water-taxis.

As documented by [Melkote and Daskin \(2001\)](#), the existing literature has pointed out the potential reduction of total travel distance with extra facilities. Nevertheless, the additional terminals in our model also lead to some new features, for instance non-loop routes and fleet redistribution, which are not addressed in the literature and will be further explained below. Without a numerical test, the potential value of the additional terminals together with other benefits will remain vague and we will not know if the saving is substantial or insignificant, in particular for the Bergen case discussed in this paper. Investment decisions in new technology, such as autonomous vessels, amount to weighing all the pros and cons on a balance scale. It is important for the decision makers to have a more elaborated understanding about different advantages and disadvantages of the technology, rather than just a rough picture, to facilitate a legitimate decision in the future. Therefore, we argue that it is necessary and valuable to discuss and test the operation of autonomous water-taxis with different number of terminals in the later computation study as it offers valuable insights into the advantages of autonomous water-taxis. Although the cost of a parking terminal for autonomous water-taxis is much lower, certain facilities, such as charging or refuelling facilities and monitoring systems, are still needed. Hence, we still distinguish between a parking terminal and a temporary docking point in this case.

Last, as mentioned before, non-loop routes can be used in the routing plan for the autonomous water-taxi, which has the potential to reduce the total travelling distance. The terms loop route and non-loop route here are mainly used to distinguish routes using the same terminal as origin and destination and routes starting and finishing at different terminals. Moreover, multiple terminals and non-loop routes also bring the company dynamic positioning capabilities so that it can redistribute the fleet among the same group of terminals or relocate the water-taxis to other terminals when demand changes over time. The possibility of parking at different terminals with non-loop routes may not be important for taxis on land, but it is critical for water-taxis. Since a water-taxi cannot park arbitrarily, it will return to terminals much more frequently. With multiple terminals, the water-taxi will have a better starting point for the next customer, which reduces the travelling distance. For example, assume that the company has two terminals, say Terminal A and Terminal B, opened in the morning with different number of water-taxis deployed. After the morning demands the company may decrease the number of water-taxis parked at Terminal A and redistribute these water-taxis to Terminal B if the demand forecast shows that more passengers will need the service near Terminal B in the afternoon. The company may even shut down Terminal A after finishing all the morning demands while opening Terminal C and relocate part of the water-taxi fleet to this newly opened terminal if the afternoon demands are neither close to Terminal A nor to Terminal B, but gathered around Terminal C.

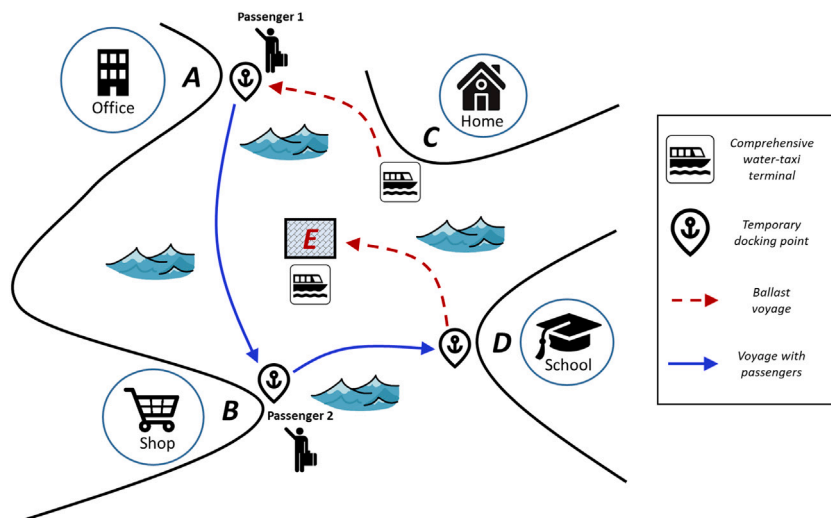


Fig. 4. Illustration of feasible solution with autonomous water-taxi.

Table 1
Summary of potential difference between conventional and autonomous water-taxis.

	Conventional water-taxi	Autonomous water-taxi
Terminal cost	High	Low
Location of terminal	Shore-based	Shore-based & Remote
Number of terminal	Single	Multiple
Routing	Loop	Loop & Non-loop
Dynamic positioning	No	Yes
Difficulty to solve	Low	High

Therefore, the daily operations of autonomous water-taxi is actually a facility location, fleet allocation and routing problem which is more complex than the case with conventional vessels (simple facility location problem only).

Fig. 4 illustrates an example for the case of autonomous water-taxi with two parking terminals. One of the parking terminals is located in a remote location, maybe a small island, while the other one sits near the residential area. The other three locations can only be accessed with temporary parking. Again, two passengers need to be transported. But this time, a non-loop route is selected due to the existence of multiple parking terminals. The water-taxi finally repositions itself at Point E, rather than returning to Point C, since based on the order information, the next customer for instance will depart from Point B after 20 min and the new remote terminal is closer to the starting point.

To summarize, the main operational difference between autonomous and conventional water-taxis can be found in Table 1. Certainly, there are other tactical and strategic factors, such as the costs of water-taxis and water-taxi terminals, that need to be taken into account when we minimize the total cost in general. We know that an autonomous water-taxi will be more expensive than a manned one due to the investment in new technology. On the other hand, a terminal serving the conventional water-taxi and its driver will also cost more. However, to the best of our knowledge, the exact cost differences between the autonomous and conventional water-taxis or terminals remain unclear. Therefore, rather than adopting certain unreliable estimations which may badly affect the results, we exclude these uncertain factors and make several assumptions for the modelling in the later section. First, the water-taxi fleet contains either autonomous vessels or conventional vessels. The taxi company does not run a mixed fleet. The number of water-taxi is given as an input and remains the same in both cases so that the comparison between autonomous and manned water-taxis are not affected by the fleet size. Second, the cost of building a comprehensive terminal for the manned water-taxi is very high so that the taxi company will invest only in one such terminal for economic reasons if conventional vessels are used. But multiple parking terminals for autonomous water-taxi are allowed since the cost is much lower.

With these assumptions, this paper will focus on the pure operational changes brought by the autonomy of the water-taxis and the corresponding savings on travelling distance. Strategic decision are included in the model as operational advantages of autonomy can only become reality if the strategic decisions fit the operational possibilities. Therefore, when this paper shows that the adoption of autonomous water-taxi leads to a shorter travelling distance, it merely means an advantage on the operational level rather than superiority in general since the tactical and strategic considerations, such as investment in autonomous technology and cost of terminals, are excluded.

The main restriction of our problem setting is the assumption of deterministic transport demand while actual demand obviously is stochastic. Normally, a deterministic model may lead to an optimistic result due to the certainty assumed. However, we argue that such a restriction will not affect the results of this paper in any major way for the following reasons. First of all, the purpose

of this paper is to seek managerial insights through the comparison between conventional and autonomous water-taxis, rather than determining a precise optimal solution for a real world application. Therefore, as long as the deterministic assumption applies to both the case with conventional water-taxis and the case with autonomous ones, the comparison remains fair. Second, although the demand in reality is stochastic, the pattern of demand during the day is predictable. For example, it is safe to assume that most passengers in the morning travel from residential areas to office areas, while in the afternoon the opposite situation can be expected. Such certainty in the transport demand pattern also guarantees that the deterministic assumption, namely the expected value of transport demand, can reflect the reality to a large extent and is good enough for the purpose of this study. Last, the pre-booking requirement also allows the water-taxi company to have certain knowledge about the transport demand in the near future. Hence, it is possible to make short-term routing decisions for the water-taxi fleet based on confirmed orders, which means a deterministic setting.

Last, we assume on-time departure as a mandatory requirement in this study, while in reality the time window constraint for a taxi service is normally soft. Such a restriction may decrease the flexibility in taxi routing and therefore slightly decrease the operational efficiency of the water-taxis in the model and lead to a pessimistic result. However, since it is also applied to both cases, the comparison remains fair and unbiased. Moreover, the pessimistic effect brought by the hard time window constraint and the optimistic effect caused by the deterministic setting will surely offset each other to a certain extent.

3. Mathematical model

To illustrate the research problem and prepare for the later computational study, a mixed-integer linear programming model is developed. The objective of the model is to minimize the total travelling distance while serving the given demands. The basic logic of the decision-making in this model can be described as follows. First, the location of the water-taxi terminal needs to be decided. For the scenario with conventional water-taxis, the company can only choose one location due to economic concerns, while for the scenario with autonomous water-taxis, multiple terminals can be opened since the cost for a terminal becomes much lower. Second, the taxi company needs to allocate its fleet to the selected terminals. This decision only applies to the case with autonomous vessels since we only have one terminal if manned vessels are used. Last, routing decisions are made based on the actual demands, terminal location and fleet allocation plan. With a fixed fleet size, the allocation plan decides the specific number of water-taxis assigned to different terminals at the start of different time periods. The allocation plan is achieved through the routing decisions in the previous time period. The purpose is to match the distribution of the water-taxis at different locations with the changing demand so as to minimize the travelling distance.

Three types of basic route are predefined in this paper. The first type connects two points with a single leg (one service voyage and no ballast voyage), the second type contains three points and two legs (one service voyage and one ballast voyage) and finally the third type involves four points and three legs (one service voyage and two ballast voyages). The three defined types are the most basic routes that are possible for serving one customer. In the model, the taxi may serve multiple customers, if their departing times fit, before the taxi returns to the depot. Such a long route then can be decomposed into the three types of basic routes which will be selected by the optimization model. For example, the route illustrated in Fig. 4 can be considered a combination of two basic routes, namely route C–A–B and route B–D–E. In general, defining three types of basic routes is mainly for modelling purposes.

The notation used in the formulation is as follows:

Sets

I	Set of locations
J	Set of routes
K	Set of demands
T	Set of time slots

Parameters

C_j	Travelling distance of route j
N_{jk}	Equals 1 if route j can serve demand k , equals 0 otherwise
D_{kt}	Number of water-taxis needed for demand k starting at time t
P_{ij}	Equals 1 if route j starts from location i , equals 0 otherwise
Q_{ij}	Equals 1 if route j ends at location i , equals 0 otherwise
E_j	Travelling time of route j
F_i	Parking capacity limitation of potential terminal at location i
G_{jk}	Ballast time needed for the water-taxi to arrive the pick-up point if route j is used to serve demand k
A_{it}	Equals 1 if a passenger departs from location i at time t , equals 0 otherwise
B_{it}	Equals 1 if a passenger will depart from location i shortly (e.g., less than 10 min) after time t , equals 0 otherwise
M	Total number of water-taxis in the fleet
L	Maximal number of terminals opened
H	Arbitrary large number for modelling purpose
ε	Arbitrary small number for modelling purpose

Decision variables

x_{jkt}	Number of times that route j is started at time t for demand k , integer variable
w_{jkt}	Number of times that route j is finished at time t for demand k , integer variable
y_{it}	Equals 1 if location i is used as terminal at time t , equals 0 if location i is used as temporary docking spot
z_{it}	Number of taxis parked at location i at the beginning of time t , integer variable
α_{it}	Equals 1 if water-taxi can park at location i (including both long-term and temporary parking) at time t , equals 0 otherwise

Objective function

$$\min \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} C_j x_{jkt} \quad (1)$$

Subject to

$$\sum_{j \in J} N_{jk} x_{jkt} - G_{jk} = D_{kt} \quad k \in K, t \in T \quad (2)$$

$$\sum_{j \in J} \sum_{k \in K} P_{ij} x_{jkt} \leq H(y_{it} + A_{it}) \quad i \in I, t \in T \quad (3)$$

$$\sum_{j \in J} \sum_{k \in K} Q_{ij} w_{jkt} \leq H(y_{it} + B_{it}) \quad i \in I, t \in T \quad (4)$$

$$x_{jkt} = w_{jk(t+E_j)} \quad j \in J, k \in K, t \in T \quad (5)$$

$$w_{jkt} = 0 \quad j \in J, k \in K, t < E_j \quad (6)$$

$$\sum_{j \in J} \sum_{k \in K} P_{ij} x_{jkt} \leq z_{it} \quad i \in I, t \in T \quad (7)$$

$$z_{i(t+1)} = z_{it} + \sum_{j \in J} \sum_{k \in K} Q_{ij} w_{jkt} - \sum_{j \in J} \sum_{k \in K} P_{ij} x_{jkt} \quad i \in I, t \in T \setminus \{t_{|T|}\} \quad (8)$$

$$\sum_{i \in I} z_{i0} \leq M \quad (9)$$

$$\sum_{i \in I} y_{it} \leq L \quad t \in T \quad (10)$$

$$z_{i0} = z_{i|T|} \quad i \in I \quad (11)$$

$$z_{it} \leq F_i \alpha_{it} \quad i \in I, t \in T \quad (12)$$

$$\alpha_{it} - 1 \leq \frac{y_{it} + A_{it} + B_{it}}{3} - \varepsilon \quad i \in I, t \in T \quad (13)$$

$$\frac{y_{it} + A_{it} + B_{it}}{3} \leq \alpha_{it} \quad i \in I, t \in T \quad (14)$$

$$x_{jkt}, w_{jkt}, z_{it} \in \mathbb{Z} \quad j \in J, k \in K, i \in I, t \in T \quad (15)$$

$$y_{it} \in \{0, 1\} \quad i \in I, t \in T \quad (16)$$

The objective function (1) minimizes the total travelling distance for serving all the demands during the entire planning period. Constraints (2) ensure that all demands are fulfilled. To be more specific, a route j satisfying two requirements will be selected for demand k at time t . The first requirement is that route j must be capable of serving demand k . The second condition requires that if ballast sailing is needed, the voyage on route j must start early enough so that the water-taxi can arrive on time and serve the customer at time t . Constraints (3) require that route j can only be initiated at time t if either its starting point (location i) is used as a terminal at time t or a passenger is departing from location i at time t . Constraints (4) make sure that route j can only be finished at time t if either its ending point (location i) is used as a terminal at time t or a passenger will depart from location i shortly (e.g., less than 10 min) after time t . Constraints (5) are bookkeeping constraints. They balance the departure and arrival of route j . If a voyage using route j starts at time t , then it also must finish at time $t + E_j$. Constraints (6) help to avoid illogical arrivals on route j . It is not allowed to have an arrival for any route at a time t which is smaller than the travelling time of the selected route. Constraints (7) ensure that the total number of voyages starting from location i in the beginning of time t must be smaller than or equal to the number of water-taxis parked at location i at that time. Constraints (8) track the number of water-taxis parked at location i at time t based on the departure and arrival in that period. Constraint (9) make sure the initial allocation of the water-taxis will not exceed the size of the given fleet. Constraints (10) restrict the number of terminals allowed to be open at time t . Constraints (11) reset the allocation of water-taxis at the end of the planning period. Constraints (12) set the maximum parking capacity for each location when parking is allowed. Constraints (13) and (14) make sure that parking at location i during time t is only allowed if location i is used as a terminal during time t (long-term parking), or a passenger is going to depart from location i shortly after time t (temporary parking). Constraints (15) and (16) define the domain of the decision variables.

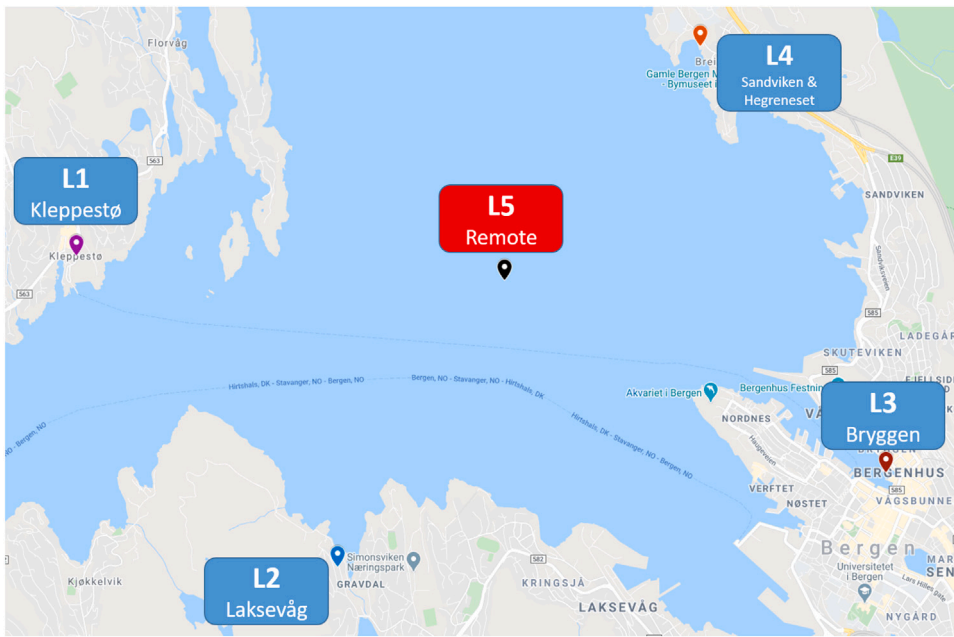


Fig. 5. Map of the test case.

As we mentioned in Section 2, this study mainly focuses on the operational changes brought by the autonomous water-taxis. The initial investment cost of the water-taxis and the corresponding terminals are not considered in the optimization model since we have very limited knowledge regarding these factors. However, the costs can be easily integrated into the model when we have more data in the future.

4. Case study

In this paper, we consider a water-taxi company transporting passengers among four locations, namely Kleppestø, Laksevåg, Bryggen and Sandviken & Hegreneset, see Fig. 5. These four locations are typical examples of areas with different characteristics in the Bergen area. Kleppestø (L1) is mainly a residential area of the municipality of Askøy nearby Bergen city. Laksevåg (L2) consists of both an industry zone and residential lands. Bryggen (L3) locates in the centre of the city. Therefore, it is a comprehensive area with offices, shops, restaurants, bars and connections to other modes of transport. Sandviken & Hegreneset (L4) are two adjacent areas with office buildings, apartments, houses and a university. Finally, the remote location (L5) is a point in the middle of the fjord with relatively equal distance to the other four locations and it is assumed to be a potential location for the terminal of the autonomous water-taxis.

Totally 180 basic routes, including 48 loop routes (starting and finishing at the same terminal) and 132 non-loop routes (starting and finishing at two different terminals), are defined to connect these five points. These basic routes are the possible options for serving a single demand. As explained in the end of Section 2, all these routes contain maximum four points and three legs (with at least one service voyage and at most two ballast voyages), for instance L2–L3–L1–L4, which is the longest possible route for one single demand. A routes covering several demands and being longer than that will be considered as a combination of multiple basic routes. In this paper, we choose to focus on the busy time of the day. Hence, the daily operating time of the water-taxi service is assumed to be 12 h which is further discretized into 144 time slots with each slot representing 5 min. The company needs to decide the locations of the parking terminals for the water-taxis, the allocation plan of the fleet and the route for each demand so that all passengers can be served on time and the total travelling distance can be minimized.

The transport demands among these locations are roughly divided into four demand waves during the day, see Table 2. A demand wave contains a group of adjacent time slots and refers to the peak hours during a day with high demand for water-taxi services. For example, the Morning Wave may refer to the time period from 6 to 9 am and consist of 36 time slots. It mostly handles people going from residential areas to working areas. The second demand wave occurs in the afternoon and serves people who need to go home after work. The third wave comes in the evening when people come out from residential areas for shopping, dining and entertainment. The last wave occurs at the end of the day for all people to go home. The detailed numbers in Table 2 refer to the total number of water-taxi voyages needed between different pairs of origin and destination during different demand waves. For example, if the Morning Wave contains 36 time slots, the total of 12 taxi calls from Kleppestø (L1) to Laksevåg (L2) are spread over these time slots in the model rather than occurring simultaneously. Table 2 shows that there are 12 individual customers (or customer groups) with different departure times, ordering water-taxi services from L1 to L2 in the morning wave. All demands are

Table 2

Transport demands among different locations during the day.

Waves of demand	Wave 1 (Morning)	Wave 2 (Afternoon)	Wave 3 (Evening)	Wave 4 (Night)
L1 – L2 (Kleppestø- Laksevåg)	12	3	2	3
L1 – L3 (Kleppestø- Bryggen)	10	3	24	3
L1 – L4 (Kleppestø- Sandviken & Hegreneset)	8	4	10	3
L2 – L1 (Laksevåg - Kleppestø)	5	17	3	3
L2 – L3 (Laksevåg - Bryggen)	11	3	3	3
L2 – L4 (Laksevåg - Sandviken & Hegreneset)	2	3	4	3
L3 – L1 (Bryggen - Kleppestø)	4	10	3	26
L3 – L2 (Bryggen - Laksevåg)	2	11	3	3
L3 – L4 (Bryggen - Sandviken & Hegreneset)	4	9	4	13
L4 – L1 (Sandviken & Hegreneset - Kleppestø)	2	5	3	6
L4 – L2 (Sandviken & Hegreneset - Laksevåg)	4	3	3	7
L4 – L3 (Sandviken & Hegreneset - Bryggen)	12	3	12	3
Total demand	76	74	74	76

Table 3

Distance among different locations.

Distance (km)	L1	L2	L3	L4	L5
L1	–	2.52	5.53	4.49	2.81
L2	2.52	–	4.17	3.99	2.02
L3	5.53	4.17	–	3.24	2.83
L4	4.49	3.99	3.24	–	1.97
L5	2.81	2.02	2.83	1.97	–

assumed to be mandatory and must be served on time in the model. Since the water-taxi service in Bergen is still a concept, no real data about the demand is available. Therefore, the demand figures in Table 2 are assumed. However, we argue that the pattern of demands at different times reflects the reality and hence the assumed data are valid for the later computational study, given the goals of this paper.

Other assumed parameters include the speed of the water-taxi, the size of the water-taxi fleet and the parking capacity of each location. First, the average speed of the water-taxi in each voyage is assumed to be 16 knots (30 km/hour). Note that the actual speed for the water-taxi in the busy area will be lower for safety reasons, while the speed may be increased in the areas with less traffic. In the Bergen case, it is not possible to avoid the low speed area through re-routing due to the port set-up. Second, we assume the taxi company has 18 water-taxis in its fleet. Furthermore, the parking capacity is assumed to differ from location to location. Location 5 has the largest parking capacity for the entire fleet due to the large space available in the middle of the water. Locations close to the city centre, on the contrary, are more likely to have limited room for water-taxi parking. Hence, we assume that both Bryggen (L3) and Sandviken & Hegreneset (L4) have a parking capacity of three water-taxis. For Kleppestø (L1) and Laksevåg (L2), not more than six water-taxis can be parked at each of the two locations. Certainly, the maximum parking capacity at different locations can be changed if the size of the terminals is changed, which may lead to a different optimal solution of terminal locations. However, the insights obtained in the later computational study will not be affected by such changes. With proper data regarding the marginal cost for extra capacity at different locations in the future, it is also possible to make the terminal parking capacity a decision variable in the optimization model.

Last, we assume the maximum time for a water-taxi to temporarily park at a docking point is ten minutes. Since only temporary parking is allowed at a docking point, a threshold has to be selected. The distances among all the points on the map are between 2 and 5 km. With the given speed, 10 min is a relatively long time. This means a water-taxi will wait at the docking point as long as the next passenger will show up within ten minutes. Otherwise, it will return to a terminal. We are aware that changes in the data will affect the specific number in the results. Nevertheless, we argue that general insights obtained through the computational study in the later section will not change because the assumptions are reasonable and realistic. The travelling distances among all locations are measured on Google Map and can be found in Table 3.

In Fig. 5, we see that the four typical locations in this case are separated by the fjord to certain degrees. For example, if a person needs to travel from Kleppestø (L1) to Bryggen (L3), it will take him around 40 min by bus and 30 min by car. But if there is a direct connection by vessel, the travelling time can be reduced to 10 min due to the shorter distance and less congestion. Therefore, great potentials can be expected for the water-based city logistics in Bergen and water-taxi is one of the alternatives. Based on the distances shown in Table 3, the authors acknowledge that the case study is based on a relatively small network with five nodes. However, this is the actual size of the Bergen port area with high potentials for water-taxi services. Moreover, the knowledge obtained based on a realistic case has value which cannot be achieved with an artificial substitute. More importantly, we argue that the insights obtained in the later computational study are independent of the size of the network since again the main purpose of this study is to explore the potential advantages of autonomous water-taxis from a pure operational perspective, rather than offer guidance for direct implementation. In the following computational study we will explore the differences between the conventional and autonomous water-taxis from a pure operational perspective.

5. Computational study

In this section, computational studies under different scenarios are conducted. The test results are then compared to obtain managerial insights. Detailed settings of each scenario are given in Section 5.1, while the results are compared and analysed in Section 5.2. The solver used in this paper is CPLEX Optimization Studio V12.6.1. The model is solved on a computer with an Intel i5-8250U 1.8 GHz CPU and 8 GB RAM. The computational time for the most complicated case (Scenario 4, explained in the later paragraph) is less than 2 h.

5.1. Tested scenarios

Four scenarios are tested in the computational study. These are “Scenario 1: Conventional water-taxi with single terminal”, “Scenario 2: Autonomous water-taxi with single terminal”, “Scenario 3: Autonomous water-taxi with multiple terminals (A)” and “Scenario 4: Autonomous water-taxi with multiple terminals (B)”. The detailed assumptions of these scenarios are as follows.

Scenario 1: Conventional water-taxi with single terminal

Scenario 1 is the benchmark scenario in the computational study. In this scenario, the water-taxi company uses conventional manned vessels. Therefore, a comprehensive water-taxi terminal becomes mandatory due to the human needs of the water-taxi drivers. Since such a terminal will be very costly, only one terminal is allowed to be built and operated in this scenario. Moreover, due to safety concerns, the remote location (L5 in Fig. 5) is not considered. Since the company can only have one terminal in this scenario, the parking capacity of this single terminal is assumed to be large enough to contain the entire fleet. Such a single terminal will then require considerable space along the shore, which will be extra costly since the shore areas are normally residential areas and have very limited space. The main decision to be made by the water-taxi company in this scenario is then the location of the comprehensive terminal so as to minimize the total travelling distance based on the given demands at different times.

Scenario 2: Autonomous water-taxi with single terminal

In Scenario 2, the fleet of water-taxis consists of autonomous vessels. Therefore, the safety concern of the taxi driver is no longer an issue, which makes it possible to use certain unprecedented locations as water-taxi terminals, for instance the remote location (L5) in Fig. 5. Except for this extra alternative location for a terminal, the rest of the settings in this scenario are the same as the ones in the benchmark scenario. We still assume that the capacity of the single terminal equals the size of the fleet.

Scenario 3: Autonomous water-taxi with multiple terminal (A)

In this scenario, we still assume that the autonomous vessels are used and the remote location is available as water-taxi terminal. Furthermore, due to the unmanned service, the infrastructures and facilities for drivers are no longer necessary. Hence, the cost of the terminal decreases dramatically, which means adopting multiple terminals becomes financially feasible. In Scenario 3, the water-taxi company can decide to build and operate a fixed number of terminals with corresponding capacity constraints and make routing decisions for its taxi fleet based on the terminal locations. The number of constructed water-taxi terminals is treated as an input rather than a decision variable since we do not know the exact cost of constructing such a terminal. Therefore, we choose to test all possibilities, namely from two to five terminals and observe the results so as to avoid potential bias caused by an inaccurate estimation of the terminal cost. Note that once a location is used as a terminal, it operates as a terminal for the entire planning period (e.g., one day) in this scenario.

Scenario 4: Autonomous water-taxi with multiple terminal (B)

The last scenario is very similar to Scenario 3. The water-taxis are assumed to be autonomous. The remote location (L5) can be used as a terminal and multiple terminals are possible. The major difference between Scenario 4 and Scenario 3 is that in Scenario 4 we assume that the water-taxi company has invested and built terminals at all five locations in the beginning. However, in order to control the operating costs of all terminals, only some of them will be opened for operation at a certain time. Although all facilities for long-term parking, such as charging and monitoring systems are turned off, the closed terminals will still serve as temporary docking points. It is possible to open different terminals at different times. For example, terminals at Kleppstø (L1) and Laksevåg (L2) may be opened in the morning while the other terminals are all closed. But in the afternoon, the water-taxi company may decide to operate terminals at Bryggen (L3) and the remote location (L5) instead. Again, the number of opened water-taxi terminal at each time is not considered as a decision variable but a given parameter because the detailed operating cost of an autonomous water-taxi terminal remains unknown. Scenario 3 is more likely to occur if the operating cost of an extra terminal, for example the increased labour needed for a shore-based monitoring centre, is negligible and lower than the potential savings on travelling distance. Meanwhile, Scenario 4 is expected if the situation is the contrary and therefore a terminal will only be open when necessary so as to save the operating cost.

5.2. Results analysis

In the computational study, we first run the optimization model with the four scenarios separately and observe the results. In the following paragraphs, the results obtained from different scenarios are compared and analysed. Managerial insights are concluded based on the comparison and analysis.

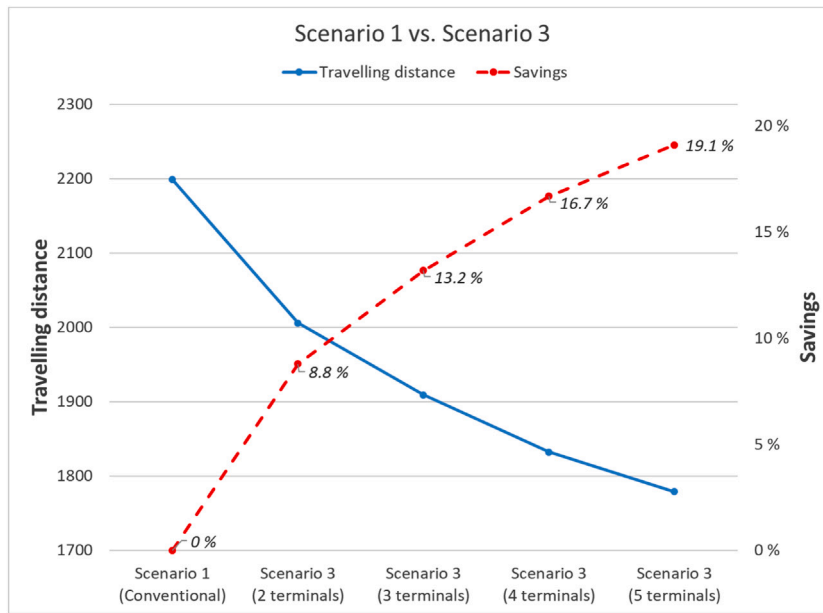


Fig. 6. Comparison between Scenario 1 and Scenario 3.

Scenario 1 vs. Scenario 2

In Scenario 1, the optimal solution suggests that the single water-taxi terminal should be built at Laksevåg (L2), while in Scenario 2 the remote location (L5) is selected as the position for the terminal. The total travelling distance needed to serve the same demands in Scenario 2 decreases by 2.4% compared to the benchmark scenario.

It is obvious that the main contributor to such a reduction in Scenario 2 is the possibility of accessing the remote location (L5) thanks to the autonomy of the water-taxi because all other factors in these two scenarios are exactly the same. Compared to the routing decisions in Scenario 1 (terminal built at L2), the routes used to serve the demands between Klepppestø (L1) and Bryggen (L3), Klepppestø (L1) and Sandviken & Hegreneset (L4), Bryggen (L3) and Sandviken & Hegreneset (L4) in Scenario 2 are substantially shorter in Scenario 2 with L5 as terminal. Although the water-taxi may slightly detour for the customers between Klepppestø (L1) and Laksevåg (L2), and Sandviken & Hegreneset (L4) and Laksevåg (L2), the total sailing distance will be reduced.

Scenario 1 vs. Scenario 3

In Scenario 3, multiple terminals are allowed to be open during the day since the cost for a terminal is much lower when autonomous water-taxis are introduced. The results in Fig. 6 show that more significant travelling distance reduction can be achieved if multiple terminals are adopted. Compared to the benchmark scenario, the savings in Scenario 3 range from 8.8% with two terminals to 19.1% with five terminals.

The greater reduction of total travelling distance in Scenario 3 (comparing to the 2.4% in Scenario 2 with autonomous water-taxis but a single terminal) is caused by the increased routing flexibility and the possibility of redistributing the taxi fleet when demand changes. For example, in the case with two terminals, location L1 and L5 are selected as water-taxi terminals while other locations only act as temporary docking points. To serve two customers in the morning demand wave, Passenger 1 from Klepppestø (L1) to Bryggen (L3) and Passenger 2 from Bryggen (L3) to Sandviken & Hegreneset (L4), a water-taxi parked at Terminal L1 the night before is assigned. A non-loop route (L1–L3–L4–L5) is adopted for this travel. The water-taxi first picks up Passenger 1 at its origin (L1), then sails to the docking point L3 to unload this customer. The vessel then parks at the docking point L3 temporarily and waits since Passenger 2 will arrive docking point L3 soon. After sending Passenger 2 to the destination (L4), the water-taxi realizes that no customer will show up at docking point L4 shortly and the vessel cannot park there for more than 10 min, since L4 is not a terminal. Hence, the water-taxi moves to Terminal L5 for long-term parking and waits for the new order. Compared to the loop routes in Scenario 1 (L2–L1–L3–L4–L2) and Scenario 2 (L5–L1–L3–L4–L5) with a single terminal, such a non-loop route substantially decreases the sailing distance for the same demands. Moreover, with multiple terminals and non-loop routes, it is possible to redistribute the fleet among the fixed water-taxi terminals in Scenario 3. For instance, by adopting non-loop routes, such as L1–L3–L4–L5, the taxi company can reduce the number of water-taxis at Terminal L1 and have more water-taxis deployed at Terminal L5 before the start of the next demand wave. Fig. 7 illustrates the redistribution of the water-taxi fleet during the day in the case with two terminals. In the beginning of the day, the company allocates 6 water-taxis at Terminal L1 and 12 at Terminal L5. Then the fleet is redistributed and 15 water-taxis are parked at Terminal L5 at the beginning of the afternoon demand wave while only 3 water-taxis are left at Terminal L1. Such a redistribution takes the demand pattern of the next demand wave into consideration and is accomplished by adopting the non-loop routes to serve the customers in the previous demand wave. Since

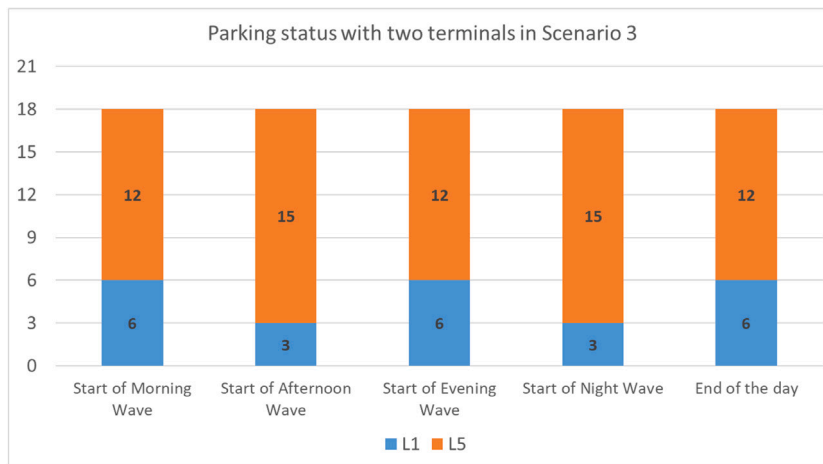


Fig. 7. Parking status with two terminals in Scenario 3.

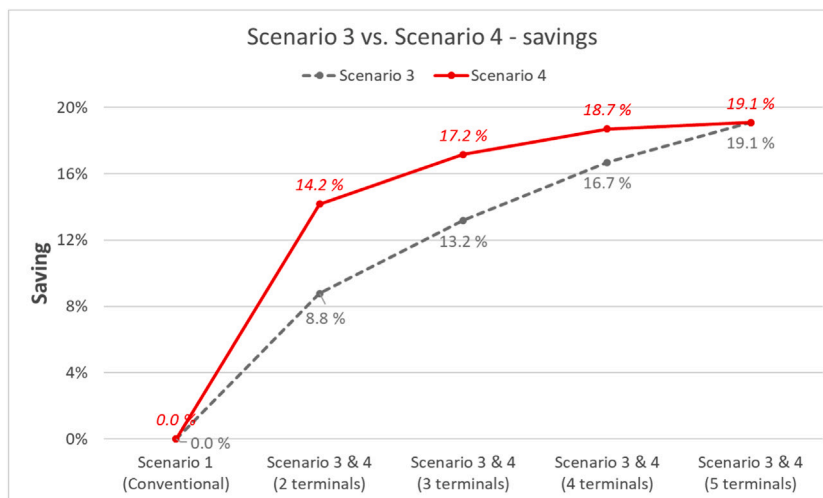


Fig. 8. Comparison between Scenario 3 and Scenario 4.

the major passenger flow in the afternoon demand wave originates from the office areas (L3, L4) and the industry zone (L2), it is reasonable to assign more water-taxis to Terminal L5 which is closer to these areas. The redistribution of the water-taxi finally leads to a better match between the fleet deployment and the demand pattern in the start of each demand wave, which helps to minimize the total travelling distance to serve the demands. Similar situation is also observed in other cases with more terminals.

Scenario 3 vs. Scenario 4

In the last comparison, we compare the results obtained in the two scenarios with multiple terminals. The main difference between Scenario 3 and Scenario 4 is that in Scenario 3 the opened terminal is decided at the beginning of the day and cannot be changed while in Scenario 4 the company may choose to open different terminals at different times. The results show that further savings on travelling distance can be expected in Scenario 4. Based on Fig. 8, we can see that the extra saving ranges from approximately 2% to 5.4% compared to the results obtained in Scenario 3.

If we compare the solutions in Scenario 4 (see Fig. 9) with the ones in Scenario 3 (see Fig. 7), we find that the extra reduction of travelling distance is brought by the relocation of the water-taxi fleet among different terminals at different times. Different from Scenario 3 in which we just redistribute the water-taxi fleet between the two fixed terminals, in Scenario 4 the water-taxis are continuously relocated to different terminals to have an even better match with the demand pattern. For example, the solutions suggest that we shut down Terminal L1 and relocate water-taxis to Terminal L3 during the afternoon and night demand waves since most passengers depart from Bryggen (L3), namely the city centre, and go home at that time. Such relocation capability brought by the autonomous water-taxi will be important when the demand pattern changes dramatically, for instance the transport demands during weekdays and weekends.

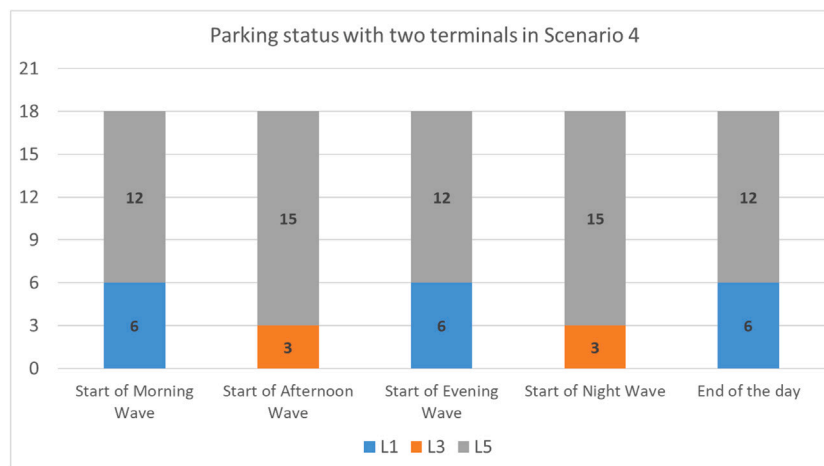


Fig. 9. Parking status with two terminals in Scenario 4.

Again, we would like to emphasize here that the insights obtained above are not only valid for Bergen or water-taxi services, but can be generalized and applied in other similar locations and applications. For example, the advantage of utilizing unprecedented locations, such as islands, as terminals for autonomous water-taxi is universal for all cities with intensive waterways. Moreover, the advantage of having multiple terminals when adopting autonomous vessels is also valid for other demand-driven waterborne transport services without fixed routes and schedules, for instance package pickup and delivery.

6. Conclusion

With the rapid development of autonomous technology, the discussion of utilizing autonomous mobility in transport and logistics systems has earned more attentions in the past few years. For countries like Norway with well connected waterways, autonomous vessels in particular has great potentials to increase the overall efficiency of the transport systems. Nevertheless, the latest survey shows that studies about the application of autonomous vessels in transport and logistics are very limited (Gu et al., 2020).

In this paper, we mainly focus on the case of introducing autonomous water-taxi in the city of Bergen. Two operational advantages of autonomous water-taxi are identified. First, without the concerns for safety and human needs of the driver, unprecedented locations, for instance an isolated peninsula or a small island, can be utilized as terminals for the parking of water-taxi. Second, operating multiple terminals becomes financially viable with autonomous water-taxi because the cost of the terminal will decrease dramatically if the facilities and infrastructures for human drivers are removed. Then a mixed-integer linear programming model is proposed to find the optimal solution for the facility location, fleet allocation and routing decisions faced by the water-taxi company during its daily operations. The model is finally used in the computational study to explore different operational strategies in the scenarios with autonomous or conventional water-taxi. The results show that besides the labour cost reduction which has been pointed out by Danish Maritime Authority (2016), significant savings on travelling distance can be accomplished if autonomous water-taxi are introduced. The reduction of sailing distance is mainly brought by the two operational advantages of autonomous vessels mentioned previously. A terminal at a remote location offers a better starting point for the water-taxi to serve the customers spread out over the entire city. Furthermore, with multiple terminals, the water-taxi company can timely redistribute and relocate its fleet among these terminals so that the allocation of taxi resources will have a better match with the demand pattern in the coming demand wave, which finally reduces the travelling distance. All these insights are also valid for other locations with intensive waterways and waterborne transport service with similar demand pattern.

However, the initial investment cost of the water-taxi fleet and the terminals are not involved in the optimization model. Therefore, the results of this paper only demonstrate the potential benefits of autonomous water-taxi at the operational level. The overall comparison between conventional and autonomous water-taxi needs further investigation with more data regarding technology costs in the future. Moreover, the case study adopted in this paper has a relatively small network and few nodes. Besides the managerial insights mentioned above, limited guidance on practical implementation of autonomous water-taxi is offered. Therefore, a more comprehensive case study with a more complex network is needed, which can be a potential future research direction.

CRedit authorship contribution statement

Yewen Gu: Conceptualization, Software, Formal analysis, Writing - original draft. **Stein W. Wallace:** Conceptualization, Resources, Writing - review & editing, Supervision.

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